Functional Requirements

1. UVUSim will read the language BasicML (signed four-digit decimal numbers) and interpret it to execute the corresponding operations.
2. UVUSim will store data and instructions in a 100 word memory with addresses starting from 00 to 99.
3. UVUSim will iterate through each instruction in memory.
4. UVUSim will have an accumulator register
5. UVUSim will have a current instruction register
6. UVUSim will process each instruction sequentially until a HALT (given by the instruction 4300) is encountered.
7. UVUSim may branch depending on the value in the accumulator if a branch instruction (40,41,or 42) is encountered.
8. UVUSim will display an error message if the input is not an integer 0-9.
9. UVUSim’s GUI has a button that allows users to open a file selection window.
10. UVUSim’s GUI has a file selection window that allows users to upload their instructions to the file if the file is in a .txt
11. UVUSim has a button to allow users to execute the instructions that are loaded in memory.
12. UVUSim will be able to write and read inputs from the user/memory.
13. UVUSim will wipe its memory and accumulator value once a new set of instructions is executed.
14. UVUSim will be able to execute basic math operations on it’s stored integers.
15. UVUSim’s GUI will allow the user to see the output of the program and enter an integer input.

Non Functional Requirements

1. UVUSim will be able to handle instructions that can be up to 100 instructions in length
2. The program will be able to run the instructions and execute them within .5 seconds.
3. The input and output of UVSim's CPU should be decoupled from the CPU's implementation. (i.e., the CPU should be able to be used with any source of input.)